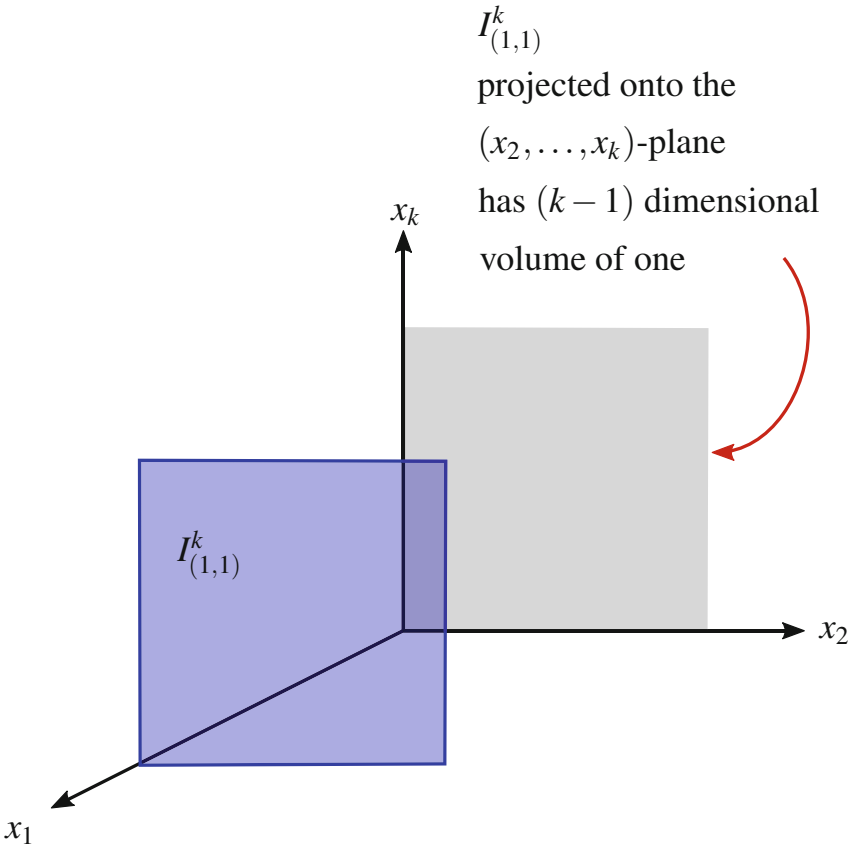
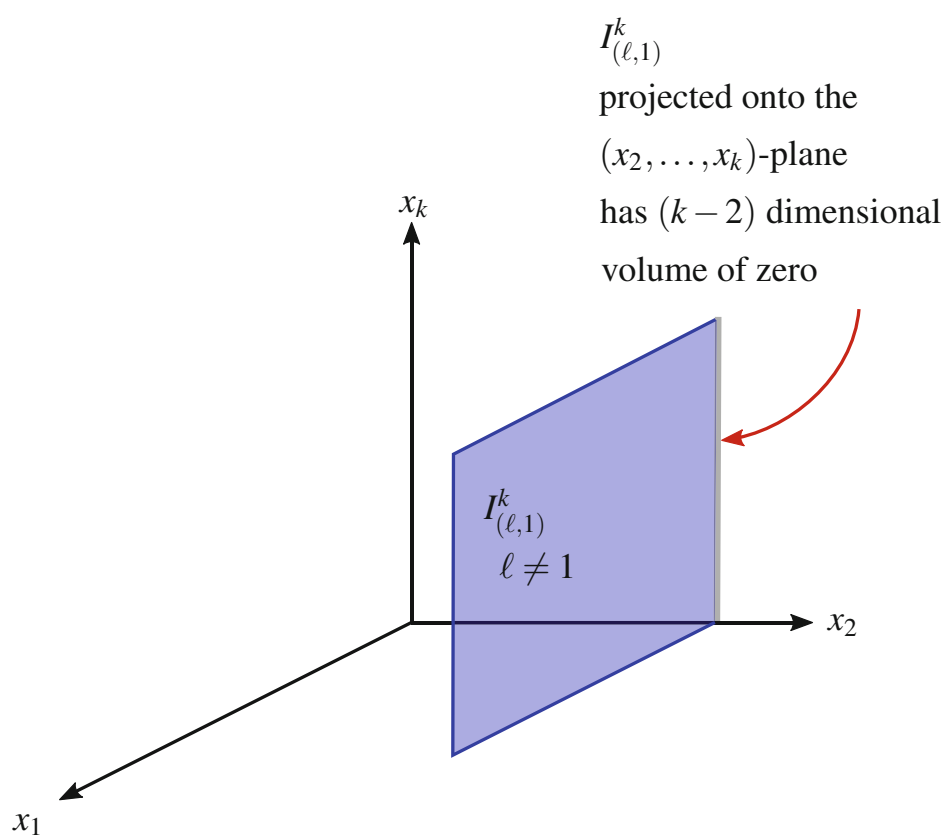


The volume form $dx_2 \wedge dx_3 \wedge \cdots \wedge dx_k$ finds the volume of the projection of $I^k_{(1,1)}$ onto the $(x_2 \cdots x_k)$-plane. The case of $I^k_{(1,0)}$ is similar. Both have a $(k-1)$-dimensional volume of one.



The volume form $dx_2 \wedge dx_3 \wedge \cdots \wedge dx_k$ finds the volume of the projection of $I^k_{(2,1)}$ onto the $(x_2 \cdots x_k)$-plane. The $(k-1)$-dimensional volume is zero. Similarly, the $(k-1)$ dimensional volumes of the projections of $I^i_{(j,a)}$ for $j = 2,\ldots,k$ and $a = 0,1$ onto the $(x_2 \cdots x_k)$-plane are zero.